

Model 15

A lo-fi channel strip in the spirit of the early-80s portable cassette multitrackers.



Overview

Model 15 is a single channel of the classic early-80s ultra-portable cassette multitracker: a low-voltage, battery-driven preamp, a fixed broad-stroke tone stage, a sluggish tape-head saturation with a top end that closes down as the level rises, and an open, breathing hiss floor with simple single-band noise reduction.

Put it on individual stems to push them towards that compressed, boxy, warm little-recorder sound. The signal runs preamp, then tone, then tape, and MIX blends the result with the dry input.

Reach for it for

- Bedroom-demo character on drums, guitars and vocals
- Warm, boxy saturation that glues transients together
- Lo-fi wobble, hiss and worn-tape dropouts
- Making a clean stem sit with genuinely lo-fi material
- Parallel tape colour with MIX

Controls

Channel

Signal light	Glows green when audio is coming out of the plugin.
GAIN	The low-voltage preamp, -6 to +24 dB (+6 dB by default). Above 0 dB it clips early and asymmetrically, smearing transients and thickening the low mids.
TREBLE	A broad high shelf at 10 kHz, -10 to +10 dB. Push it hard and the shelf itself turns a little gritty and grainy rather than staying clean.
BASS	A broad low shelf at 100 Hz, -10 to +10 dB, with the same gentle grit when pushed.
TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
MIX	The coloured signal against the dry input, 0 - 100%.

Meters

L / R	VU meters showing the left and right output.
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Tape

DRIVE	The tape saturation, 0 - 100% (35% by default): a sluggish magnetic memory ahead of a soft saturation curve, plus a top end that rolls off further as the level rises, gluing transients together. A small, fixed head bump around 80 Hz is always there.
WOW	Wow and flutter in the transport, 0 - 100 (25 by default): a slow wow, a slower random wander and a faint 6 Hz flutter.

HISS	The tape's own noise floor, 0 - 100 (50 by default). It ducks a little under the signal and swells back in the gaps. Each side has its own hiss.
AGE	An older, worn tape, 0 - 100 (15 by default): the top end goes further and short dropouts creep in, more often the higher you turn it.
OUTPUT	The output level, -24 to +12 dB.
NR	The simple B-type-style noise reduction, in by default: the quiet highs are lifted before the tape and taken back down after it, which lowers the hiss. Out gives the raw tape, hiss and all.

The header

< and >	Step to the previous or next preset.
PRESETS	Opens the list of factory presets. A preset sets GAIN, TREBLE, BASS, the TAPE controls, NR and MIX; OUTPUT, TRIM and GUARD are left as they are. The last entry resets everything to the defaults.
PATCH display	Shows the name of the preset in use.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Output stage (REAPER's parameter list only)

Model 15 has the GFX output stage, but no knobs for it on the panel. Its settings are in REAPER's parameter list, and Reset all to defaults puts them back.

Band sat / Bias	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean). Bias adds more even (2nd) harmonics. Off by default.
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Sat low band / Sat high band	Where the low and high saturation bands start (40 - 400 Hz and 2 - 16 kHz).
Ceiling / limit threshold	With Clip: the level the peaks are softly clipped at. With Limit: the threshold of a brickwall limiter that holds the peaks at -0.3 dBFS.
Output stage	Off, Clip (a soft clipper, no latency; the default) or Limit (a lookahead limiter, 3 ms latency, compensated by REAPER).

Presets

Bedroom Demo	The standard setting with a touch more TREBLE, BASS and DRIVE, NR in.
Hot and Boxy	The preamp pushed to +15 dB into 70% tape drive, with more hiss and age.
No NR, Open Tape	Noise reduction out and the hiss turned well up.
Worn-out Portable	Heavy wow, lots of age and dropouts, a little extra bass.
Clean-ish Portable	Preamp at 0 dB, light drive, wow, hiss and age.
Reset all to defaults	Every control back to its starting value, including OUTPUT, TRIM, GUARD and the output stage.

Tips

- GAIN and DRIVE do different things: GAIN gives the preamp's early, edgy clipping, DRIVE the softer tape squash and darkening. Try one high and the other low.
- On a set of stems, keep WOW low on all but one or two so the mix doesn't drift out of tune with itself.
- Switching NR out is the quickest way to that hissy, open home-recording sound.
- Turn MIX down to 40 - 60% to keep the attack of drums while adding the tape's body.

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